Machine Learning Techniques for Short Term Stock Movements Classification for Moroccan Stock Exchange

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***Abstract*—Accurate stock price forecasting is important for investors and traders to make informed trading decision. However, prices have a complex behavior due to their nonlinearity and nonstationarity. In this paper three Machine learning techniques are implemented to predict a very short term (10 minutes ahead) variations of the Moroccan stock market: Random Forest (RF), Gradient Boosted Trees (GBT) and Support Vector Machine (SVM). A selection of technical indicators was used as inputs variables and a feature selection and samples selection steps were performed to improve prediction accuracy and training time. An eight-year period of intraday prices (tick-by-tick data) of Maroc Telecom (IAM) stocks is employed as experimental database to evaluate the performances of the selected models. The experimental results have shown that RF and GBT are superior to SVM for our dataset. Further, the low computational complexity and reduced training time of RF and GBT are suitable for short term forecasting.**

Keywords—Random Forest; Gradient Boosted Trees; Support Vector Machine; Stock Market Prediction

I. INTRODUCTION

Stock market prediction is undoubtedly one of the most challenging issues for traders and researchers. The opportunities of high profit offered by the stock market have always attracted a large number of investors. However, this profit is strongly associated with a greater risk of loss which is caused by the fluctuations of the market ; hence the importance of accurate prediction of market behavor in order to make informed trading decision. In the past the majority of forecasts were related to the change in price for one day ahead, but with the volatility of markets and the growth of electronic trading platforms this forecasting horizon has been reduced to become few minutes to few seconds ahead.

Researchers consider stock market prediction as a challenging task due to the difficulty in capturing the non-linear and non-stationary variation in data. The application of machine learning techniques for stock market prediction is a well established area. But for the Moroccan stock market, there is a big lack regarding the use of these techniques.

In this work, our goal is the classification of the direction of variation (Up, Down) of the IAM stock price in the very short term (10 min ahead). Three techniques will be implemented and tested: Random Forest, Gradient Boosted Trees and Support Vector Machine. Classification is based on a set of technical indicators computed from tick-by-tick data. The comparison of these algorithms is performed by the analysis of the accuracy and the CPU Time.

The remainder of this paper is organized as follows. Section II deals with the literature review. A brief presentation of RF, GBT and SVM is outlined in section III. Details about experimental setup can be found in section IV. The experimental results are presented in section V. Finally, we conclude this study and discuss possible future work.

II. LITERATURE REVIEW

Stock market is a highly volatile and changing environment ; hence the difficulty to predict its future trends and movements with high accuracy. In the past, stock market forecasting was a question of intuition and feeling of traders. As the market and trading volumes grew, more sophisticated tools and models are used to perform more accurate and robust predictions. Fundamental analysis, time series analysis, technical analysis and machine learning techniques are used, among other techniques, in this area.

Fundamental analysts believe that market is 90 percent logical and only 10 percent psychological [1].This type of analysis states that the value of a stock is determined by its potential profitability thus by analyzing the factors that influence the profitability of a company it identifies whether a stock is under valuated or overvalued [2]. Due to the nature of the used data (Macroeconomic indicators, Industry analysis, Company analysis …) fundamental analysis is useful when doing long term forecasting.

Time series models are widely used in stock market prediction due to simplicity to develop and implement them. A large number of models can be found in the literature (ARIMA, ARCH/GARCH...). Time series models are about analyzing historic data and attempting to approximate future values of a time series as a linear combination of these historic data [3].

Technical analysis is the study of stocks prices, with charts being the primary tool [4]. Price and volume are the building blocks of the technical indicators that are computed to capture the signals of future changes.

In recent years, machine learning techniques have been widely applied for predicting stock market:

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Artificial Neural Network (ANN) represents one widely used machine learning technique for stock market forecasting. ANN is a computing system with nonlinear computing unit interconnected by links [4] it is an excellent forecasting tool which can learn by adjusting the interconnection between layers [5]. ANN has very interesting properties: learn highly non-linear models, can handle noisy data, and can use inputs of different kind [6]. These characteristic make them useful for forecasting financial time series. However, ANN suffers from: (i) Local Minimum Traps (ii) difficulty in determining the hidden layer size and (iii) learning rate [7]. To overcome these weaknesses, researchers use ANN as a baseline model and "hybridize" it with others methods: Fuzzy Inference System [5] [8], Rough Set (RS) [9], Ensemble of classifiers [10] [11], Genetic Programming [12], GARCH models [13] [14], Evolutionary algorithms [15], Hidden Markov Model (HMM) and Genetic Algorithms (GA) [16], Dynamic time windows and Case Based Reasoning (CBR) [6], Self-Organizing Map [17].

Support Vector Machines (SVM) has been widely applied to build classifiers for stock markets. Unlike ANN, SVM has excellent generalization propriety. It embodies Structural Risk Minimization principal (SRM) [18]. To enhance the performance of SVM in stock market forecasting researchers used different techniques based on parameter optimization and Ensemble Classifiers:

Parameter optimization: Adaptive parameters [19], grid-search technique [20][21][22][23], swarm optimization [24], Conjugate gradient method [19], sequential minimal optimization, gradient projection method [25], different parameters combination [26] [27] [28].

Ensemble classifiers: Bagging, AdaBoost [11], Multiple-kernel learning [25] [23].

Decision Tree (DT) is one of the most used algorithms by researcher for the forecasting of financial time series because it creates rules wish are easy to understand and interpret [10]. The following forms of DT are used in market prediction: Classification and Regressing Tree (CART) [10] [11], Decision tree generated by a Genetic Algorithm [29], Random forest [30]

III. A REVIEW OF RANDOM FOREST, GRADIENT BOOSTED TREES AND SVM

In this paper, for the purpose of classification of Moroccan Stock Market movements we will use three well known techniques of machine learning: Random Forest, Gradient Boosted Trees and Support Vector Machine. The choice of these models is based on their prediction capability, low computational complexity and reduced training time. In this section the basic concepts of these models are briefly described.

A. Random Forest

Random Forest (RF) is an ensemble learning method for both classification and regression problems [31]. RF is a collection of decision trees that grow in randomly selected subspaces of feature space. It employs Bagging method to produce a randomly sampled set of training data for each of the trees and gives a prediction based on the majority voting (classification) or averaging (regression).

Unlike CART model which performs a growing step followed by a pruning one, random forest building strategy is as follows [32]:

- At each node, a given number of input variables are randomly chosen and the best split is calculated only within this subset
- No pruning step is performed, so all the trees of the forest are maximal trees

The bagging method used by RF improves the stability and accuracy of the model, reduces variance and helps to avoid overfitting [33].The two main parameters of RF are: the number of trees in the forest and the number of input variables randomly chosen at each split.

B. Gradient Boosted trees

The Gradient boosting technique is a technique used for regression and classification problems. It works by producing an ensemble of weak classifiers. The Gradient Boosting Algorithm works as follows [34]:

For a training set (x_1, y_1), (x_2, y_2),…,(x_n, y_n) and a differentiable loss function $G(y, f(x))$ and the iteration number M.

Initialize $f_0 = \arg min_\beta \sum_{i=1}^{n} G(y_i, \beta)$

For $p = 1\ to\ M$ compute:

$$Z_{ip} = -\left[\frac{\partial G(y_i, f(x))}{\partial f(x_i)}\right]_{f(x)=f_{p-1}(x)} \qquad i = 1, 2, \dots . n$$

Let N be the number of the leaves of a decision tree. The tree divides the input space into N regions: r_{1p} , r_{2p} ,…, r_{Np}

Fit a basic leaner $h_p(x)$ to (x_1, z_{1p}), (x_2, z_{2p}),…, (x_n, z_{np}) :

$$h_p(x) = \sum_{j=1}^{N} a_{jp} I(x \in r_{jp})$$

Where a_{jp} is the value predicted in the regionr_{jp} .

$$f_p(x) = f_{p-1}(x) + \sum_{j=1}^{N} \beta_{jp} h_p(x) I(x \in r_{jp})$$

Where: $\beta_{jp} = \arg min_\beta \sum_{x \in r_{jp}} G(y, f_{p-1}(x) + \beta h_p(x_i))$

For the number of terminal leaves to be considered, it is recommended to use values in the range of $4 \leq N \leq 8$. N beyond this range is not recommanded

C. Support Vector Machine

SVM[18] are used for classification and regression analysis. SVM benefits from two ideas [35]: maximizing the margin and the kernel tricks. It works by constructing an optimal separating hyperplane with the maximum margin (green line on Fig 1). SVM maps the original input feature space into a higher dimension so that it can be separated by a linear model in the higher dimensional space (hard margin), the linear, RBF and sigmoid kernels can be used for this purpuse. The input vectors that define the width of maximum margin are called Support Vector [36].

For the case of non-separable feature space, a soft margin is applied to allow some points to be misclassified. A penalty parameter C (C>0)can be set to be the upper bound in order to control the amount of deviation to be tolerated. In classification, SVM solves the quadratic optimization problem in (1): Where n is the number of training examples and ξ_i are positive slack variables.

$$\min \| w^2 \| + C \sum_{i=1\ldots n} \xi_i \qquad (1)$$

$$s.t. (\{w, x_i\} + b) y_i > 1 - \xi_i , \xi_i \geq 0, \quad i = 1 \ldots n$$

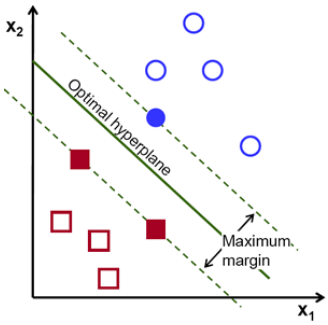

Fig 1: Geometric description of the SVM margin [37]

IV. EXPERIMENTAL SETUP

In this section we describe the different experimental setup adapted in this work.

The fig 2 shows the main steps of the Short Term Stock Prediction Process adopted in this work: The Data Preprocessing step consist of raw data gathering and transformation to input features. The modeling step deals with parameter optimization of the models. The Testing step assesses the accuracy of different models.

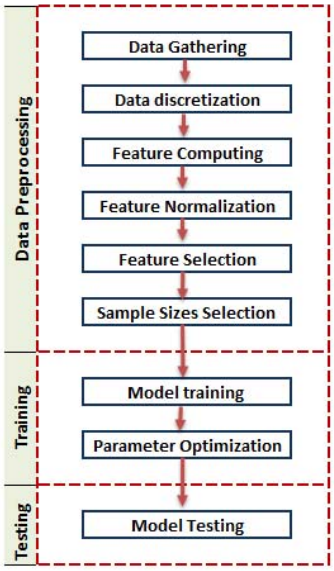

Fig 2: Short Term Stock Prediction Process

A. Input Data

In this work we use intraday data of IAM stocks prices as experimental data.

The Moroccan Casablanca Stock Exchange (CSE) is one of the emerging stock markets in Middle East and North Africa (MENA). It was established in 1929 and it is currently under a big regulatory and technology reform. It has 75 listed companies with a total market capitalization of $50 billion in june 2016 [38].

Maroc Telecom (IAM): is a listed company in the Casablanca stock market [38], it provides various telecommunication services in Morocco and internationally. IAM is one of the biggest capitalizations in the CSE ($10 billion in June 2016 (20% Market Capitalization)) it is also one of the most liquid and most traded securities in this market.

The data used in our study are intraday data (tick-by-tick data). This data record the trades executed for a stock in each day and are characterized generally by a small time span (TABLE 1). The whole data set covers the period from 04-03-2008 to 31-12-2015.

TABLE 1: Example of tick-by-tick data

Stock	Date	Time	Price (MAD)	Quantity
IAM	04/03/2008	10:45:07	186	303
IAM	04/03/2008	10:45:07	186	1097
IAM	04/03/2008	11:02:42	185,5	20

B. Data Preprocessing

a. Target variable:

Trades data does not arrive at regular intervals which makes it difficult to build classifiers. To escape the irregularly spaced tick-by-tick observations we have transformed our data to regularly spaced intervals by 10 minutes. For each interval, we consider the price of the last recorded transaction or the price of the previous interval in the absence of transactions in the current interval (Fig 3).

Fig 3: Trades (transactions) arrival at irregular time

The choice of the intervals of 10 minutes was based on the analysis of the transaction occurrence frequency : on average the time between the occurrence of tow trades is 10 minutes. After discretization our database contains: 18 001 observations/ price variations (Up, Down).

For predicting the direction of change of IAM stock prices, we adopt the following codification for the target variable:

Up= if the price has increased between two successive intervals

Down= if the price has decreased between two successive intervals

b. Choice of inputs variables:

Appropriate input variables need to be chosen to predict the direction of market movements in the short term with high accuracy. We will test two types of input variables: raw lagged values and technical indicators.

1) Lagged values:

To assess the predictive power of lagged price series as input variables we consider two sets of lagged variables which we feed to a random forest to assess their predictive power:

- **20 lagged values** as input variables give a result of 47% of OOB error.
- **110 lagged values** as input variables give a result of 51% of OOB error.

We can conclude that the use of raw lagged variables as input variables has no significant added value.

2) Technical indicators

Markets analysts use a big number of technical indicators (TI). For the sake of simplicity, we have selected 23 among the most frequently used technical indicators. We use these TI as input variables. Basically, these indicators are computed on daily data but to use them in intraday basis they are computed over the 10min intervals (TABLE 2).

The training of a random forest with this variable gives 19% of OOB error. This result shows that the use of technical indicators as input variables can lead to good result.

TABLE 2: list of Technical Indicators

Variables	Description
MA5, MA3, MA10, EMA5, EMA3, EMA10	**MA[q]:** is the average price of a stock over q periods. **EMA[q]:** similar to MA but assigns weight to more recent prices.
P(t-1), P(t-2), P(t-3), P(t-4), P(t-5)	5 lagged values
ROC(1), ROC(2), ROC(3), ROC(4), ROC(5)	5 Rates of change
SD10	Standard Deviation over 10 last intervals
PSL10	Psychological line over 10 last intervals
%K50, %D50	Stochastic oscillator: it contains two lines (%K and %D)
RSI10	Relative strength index line
Indicator Variable =(1 or 0)	Indicates if the previous interval was un "Up" =1 or "Down" =0 (Boolean)

Since our goal is classification of market movements in the short term, these technical indicators can be computed in parallel : after each 10min the TI are computed to be fed to the models as input data to make a classification for the next 10min interval.

c. Input variables transformation:

Particular attention should be paid to the scale of data to be used. We use the normalization technique to transform our variables. The normalization transformation maps all variables values to the same range (e.g. [0, 1]).

C. Choice of sample size to use

The sample size to be considered must be evaluated via the performance of a model on different databases with different sizes. The whole data set is divided into 5 subsets (Fig 4). Each subset is divided into two parts. The first part (2/3 of observations) is used to train the models. The second (1/3) part is for out-of-sample evaluation and comparison of performances of different estimated models

To select the best sample size to use we train a random forest with the 5 subsets (TABLE 3). The use of RF for this task is mainly motivated by its relative ease of use.

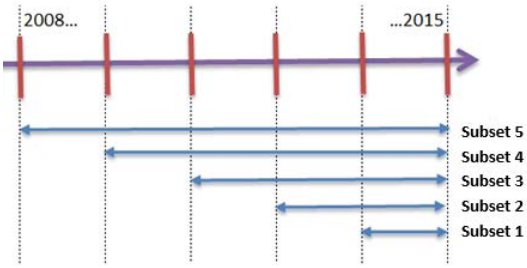

Fig 4: Selection of the five subsets

TABLE 3: Samples sizes

	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5
Training	2 880	5 760	8 640	11 521	12 001
Test	720	1 440	2 160	2 880	6 000
Testing Error	**4.8%**	**6.7%**	**8.1%**	**10.7%**	**18.8%**

It is clear from (TABLE 3) that the use of the latest data gives better results and the use of older data affects the accuracy of the model. This result seems consistent with the context of our forecasting horizon: we seek to make a prediction in the short term; the most recent data should have more impact than the older ones.

In the following sections we will use only the newest data (dataset 1) to train our models.

D. Feature selection

Before applying a machine learning technique, a feature selection step has to be performed to select features that describe the events to be modeled [10]. Feature selection step helps: decreasing the noise in dataset; reducing the computational cost; better understand the final models [24].

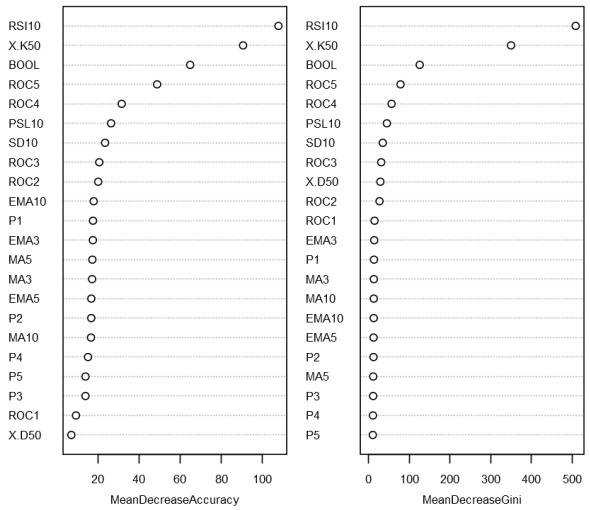

Fig 5: Plots of variables importance

Ignoring the feature selection problem can have negative effects such as over fitting and is likely to damage overall results [28].

We use Random Forest model to assess the importance of each input variable. RF provide two methods for feature selection: mean decrease impurity (Gini impurity: for classification) and mean decrease accuracy. Based on the variable importance plot on (fig. 5) we can see that the following variables are the most important ones: BOOL, ROC(2), ROC(3), ROC(4), ROC(5), SD10, PSL10, %K50, %D50, RSI10.

V. EXPERIMENTAL RESULTS

In this section we will present the results of our different models.

A. Random Forests

We train a random forest with the 10 input variables; the random forest is constituted with 500 trees and 7 input variables are randomly chosen at each split. On the test set we have got a testing accuracy of **95%.**

B. Gradient Boosted trees

The trees in our Gradient Boosted Trees model have 6 terminal leaves. The estimation of the model has needed 61 iterations. For the Gradient Boosted Trees model we have got a testing accuracy of **94%.**

C. Support Vector Machine

In this section we will explore the predictability of the (up, down) movements of IAM's price with SVM by using different Kernel functions.

a. Linear kernel

The linear kernel SVM has a parameter to be determined: the penalty factor "C". After trying multiple values for the parameter C we find that the value C=1 gives the best testing accuracy of **89%.**

b. Radial kernel

The model selection in the Radial SVM algorithm is based on two input parameters: the RBF kernel parameter "γ" and the penalty factor "C". We use the hold-out procedure (training/testing) to determine these parameters. We use the grid search method to find the optimal combination of these inputs parameters.

The best combination (γ, C) found by the grid search is: $\gamma=2^{-5}$ and $C=2^{6}$. The testing accuracy of the estimated model is: **90%**.

c. Sigmoid kernel

Sigmoid SVM has an extra parameter to be tuned in addition to the parameters (γ, C). But in our case for the sake of simplicity we decided to optimize only the parameters (γ, C).

The best combination (γ, C) found by the grid search is: $\gamma=2^{-6}$ and $C=2^{7}$. The testing accuracy of the estimated model is: **88%.**

The relative low performance of the SVM models can be explained by the choice of the values of hyper-parameters (γ, C). We have used grid search method to find the optimal combination of these parameters other methods could lead to better result.

D. CPU time analysis

Forecasting future market movements within short time require the use of fast models. A model that needs 15 minutes to make a prediction is useless when making forecasting within 10 min ahead ; hence the need to analyze the CPU times of the used models.

TABLE 4 shows the CPU time for the training step (parameters tuning + parameters estimation): as we can see all the tested models have a reasonable response time so they are useable within the forecasting horizon of 10 min.

TABLE 4: CPU Time of the three models

Models	CPU Time (s)
RF	3
GBT	0.88
Linear SVM	0.53
Radial SVM	217
Sigmoid SVM	318

VI. CONCLUDING REMARKS

In this study, a comparison between Random Forest, Gradient Boosted Trees and SVM for forecasting Moroccan stock market is performed. The empirical results show that all the models give very satisfactory results and have a small response times, making them useable for short term forecasting. The experimental results have shown that RF and GBT are superior to SVM for our dataset. As future works we suggest: The use of other input variables; the use of more

sophisticated feature selection techniques; the use of adaptive techniques that can handle data structure change; the use of three classes: Up, Down and Unchanged and the use of other evaluation strategies.

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